

PROVISIONAL PATENT APPLICATION

**REPRESENTATION IMPROVEMENT AND
GENERATOR COMPRESSION SYSTEMS AND
METHODS**

Inventor(s)	Mark White
Applicant / Assignee	Forma Substrate Inc
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Abstract

Systems and computer-implemented methods are disclosed for improving computational representations by generating or receiving multiple candidate representations of common or related observations, measuring structural organization exposed by each candidate, identifying compact generative structure associated with the observed organization, and selecting or proposing a representation according to explicit structural, predictive, complexity, qualification, or operational criteria. A representation-improvement engine may construct candidate views using different transformation rules, constraints, feature families, recursive indexes, scales, graph or transport structures, or generator definitions. A structural-analysis engine measures one or more of neighborhood stability, basin organization, path accessibility, transport, bottlenecks, recursive stability, structural regimes, recovery behavior, compression, or other organizational properties. A generator-compression subsystem seeks a smaller or simpler generative organization that reproduces or preserves selected structural relationships or future possibilities. Candidate representations may be compared only after independent qualification or may be ranked provisionally during engineering and later submitted for independent evaluation. In some embodiments, a plurality of scientifically entitled candidates forms an entitlement region, and an engineering policy selects among the entitled candidates according to complexity, computational cost, latency, interpretability, predictive utility, robustness, or another explicit criterion. The process may iterate by using evidence from one cycle to propose a subsequent representation, thereby treating representation improvement as a persistent computational capability rather than a one-time artifact. Autonomous unconstrained generator discovery is not required; early embodiments may operate over curated, parameterized, or composable generator families.

1. Technical Field

The present disclosure relates generally to computer-implemented representation learning, scientific computing, machine learning, structural analysis, recursive indexing, model selection, graph and topological computation, dimensionality reduction, compression, scientific discovery systems, and decision support.

More particularly, the disclosure relates to systems and methods for generating, comparing, compressing, qualifying, selecting, and iteratively improving computational representations according to the structural organization and generator structure they expose.

2. Background

Computational systems ordinarily assume a representation, optimize within that representation, and evaluate performance against a task objective. The representation itself is often treated as a fixed input or as an intermediate learned artifact rather than as an object that can be systematically improved.

Two representations can encode the same underlying observations while exposing materially different neighborhoods, transitions, pathways, bottlenecks, recursive hierarchy, regime boundaries, or extrapolation behavior. Accordingly, the quality of a downstream prediction or decision can depend on what relationships the representation makes visible.

Conventional compression generally asks how to reduce storage, dimensions, parameters, or bits while preserving reconstruction or task performance. The inventors recognized a different form of compression: reducing observed organizational complexity into a smaller generative description that preserves or explains important relationships, transformations, pathways, or future possibilities. This is referred to herein as generator compression.

Conventional model-selection procedures often select a model or embedding from a predefined set based on a scalar objective. Such procedures do not necessarily ask whether an alternate representation exposes more

stable or useful structure, whether multiple representations are scientifically equivalent within evidence tolerance, or whether a simpler generator can reproduce the organization that matters.

A need therefore remains for an architecture in which representation construction is itself iterative: candidate representations are generated, their structural organization is measured, candidate generator structure is compressed or compared, evidence is accumulated, and a subsequent representation is selected or proposed under explicit scientific and engineering rules.

The inventors further recognized that representation improvement should remain separate from scientific qualification. Engineering may propose or optimize a representation, but a distinct qualification process may be required before the representation supports scientific interpretation, prediction, action, publication, or other downstream use.

3. Summary of the Disclosure

In one aspect, a representation-improvement system receives a source observation set and a plurality of representation generators or construction rules and constructs candidate representations that differ in at least one structural assumption, feature family, constraint set, scale, transformation rule, or recursive organization.

In another aspect, a structural-analysis engine measures the organization exposed by each candidate representation. Structural measurements may include stable neighborhoods, admissible relationships, recursive regions, basins, paths, transport organization, bottlenecks, structural entropy, regime stability, excursion/recovery behavior, or another relational property.

In another aspect, a generator-compression subsystem identifies a compact generative description that reproduces, approximates, preserves, or explains selected structural organization of a candidate representation. Compactness may refer to fewer generator rules, lower description length, fewer active relations, fewer parameters, lower computational cost, fewer recursive levels, or another reduction in generative complexity.

In another aspect, a representation is considered improved when it provides increased useful structural exposure, equal or improved task performance, reduced complexity, greater robustness, stronger qualification, safer abstention, or another explicitly defined advantage relative to a reference representation.

In another aspect, candidate comparison is multidimensional. A representation need not dominate every measurement. The system may preserve a Pareto set, entitlement region, or policy-defined candidate set rather than forcing all evidence into a single scalar score.

In another aspect, representation engineering and scientific qualification remain separate. Candidate representations may be generated and screened during engineering, then frozen and submitted to an independent qualification subsystem. Scientific entitlement is determined from qualification evidence rather than by the representation-improvement engine itself.

In another aspect, representation improvement is recursive. Evidence from one cycle is used to diagnose limitations, propose a new generator or representation, construct a new candidate, and repeat the comparison process.

In another aspect, early implementations use curated or parameterized generator families rather than unconstrained autonomous search. Generator families may be manually defined, selected from a registry, composed from known primitives, or varied within bounded parameter sets.

In another aspect, the system records the entire improvement lineage, including source observations, parent representation, generator definition, structural evidence, predictive evidence where used, qualification record, selection policy, and resulting child representation.

Core inventive combination: The disclosure is directed to representation improvement as a computational loop: generate candidate representation -> expose and measure organization -> compress or identify generator structure -> compare candidates under explicit evidence and complexity criteria -> qualify as required -> select or propose the next representation -> repeat. The process, rather than any one representation artifact, is the protected capability.

4. Brief Description of the Drawings

FIG. 1. Overall representation-improvement loop from observations through candidate generation, structural measurement, generator compression, comparison, selection, and subsequent representation.

FIG. 2. Example candidate-representation generator registry.

FIG. 3. Example generator-compression architecture converting observed organization into a compact generative description.

FIG. 4. Example multidimensional representation comparison profile.

FIG. 5. Example entitlement region and engineering selection from qualified candidates.

FIG. 6. Example recursive representation-improvement lineage.

FIG. 7. Example separation between representation engineering and independent qualification.

FIG. 8. Example generator-entitlement architecture linking claimed explanatory scope to demonstrated generator capacity.

FIG. 9. Example structural measurements used in representation comparison.

FIG. 10. Example implementation with a representation runtime, structural computation engine, qualification subsystem, and recursive discovery controller.

FIG. 11. Example curated generator search space with bounded composition and parameter variation.

FIG. 12. Example representation-improvement record stored with provenance and parent-child lineage.

5. Definitions and Interpretive Conventions

Term	Meaning
Representation	A computational organization of observations, states, identities, features, relationships, constraints, coordinates, neighborhoods, paths, hierarchy, topology, or derived structure.
Representation generator	A rule, function, transformation, construction procedure, parameterized operator, composition of operators, or other mechanism that produces a representation from source observations or another representation.
Generator	A representation generator or a compact rule set capable of reproducing or approximating selected

	organizational structure. The term does not require a unique causal mechanism.
Generator compression	Reduction of observed or represented organizational complexity into a smaller, simpler, lower-cost, or more compact generative description that preserves or explains selected structural relationships or future possibilities.
Representation improvement	A process of generating, comparing, qualifying, selecting, or proposing representations according to the useful structural organization, generator structure, predictive behavior, robustness, complexity, or entitlement they expose.
Structural exposure	The degree to which a representation makes a selected organizational property measurable, stable, reproducible, discriminable, or operationally useful.
Candidate representation	A representation proposed for comparison or qualification.
Reference representation	A baseline or prior representation against which another candidate is compared.
Representation lineage	The parent-child or derivation history connecting source observations, generators, candidate representations, qualification records, and selected descendants.
Generator entitlement	The degree to which available evidence supports the explanatory or predictive scope attributed to a generator or generator-based representation.
Curated generator family	A predefined set of representation generators selected by a human, engineering process, registry, library, or bounded rule system.
Representation-improvement cycle	A sequence comprising at least candidate generation, structural measurement, comparison, selection or proposal, and generation of a subsequent candidate or result.

6. Overall Representation-Improvement Architecture

Observations O
 $|$
 v

```

Candidate Generator Set G = {G1, G2, ... Gn}
|
v
Candidate Representations R1...Rn
|
v
Structural Measurement S(Ri)
|
+--> optional task / predictive evidence P(Ri)
|
v
Generator Compression / Complexity Characterization C(Gi, Ri)
|
v
Candidate Comparison
|
+--> Engineering screening
+--> Independent qualification
|
v
Entitled / supported candidate set
|
v
Selection or next representation proposal
|
v
R{t+1}
|
repeat

```

The system may terminate after one cycle, continue until a threshold is met, continue for a fixed number of cycles, or proceed until human authorization is withheld. A fully autonomous search is not required.

7. Candidate Generator Registry and Construction

An early embodiment uses a registry of known generator definitions. Each generator identifies how a candidate representation is constructed.

```

GeneratorDefinition {
  generator_id
  input_type
  output_representation_type
  transformation_rules[]
  constraint_rules[]
  scale_policy
  recursion_policy
  parameters
  dependencies[]
  cost_profile
  provenance
}

```

- Direct or coordinate representation generators.
- Constraint-derived or compatibility-weighted representation generators.
- Neighborhood or graph representation generators.
- Recursive-region or basin representation generators.

- Path or transport representation generators.
- Hierarchy or multiscale representation generators.
- Biochemical, relational, symbolic, geometric, probabilistic, or domain-specific generators.
- Compositions of known generators in which the output of one generator becomes the input of another.

7.1 Bounded variation

The registry may permit bounded variation of generator parameters without permitting unrestricted generator invention. Examples include varying neighborhood size, recursion depth, scale, weighting rule, constraint parameter, feature-family inclusion, compression level, or other predefined setting. The bounds and search policy may be frozen before confirmatory evaluation.

8. Structural Measurement of Candidate Representations

Structural category	Illustrative measurements
Neighborhood structure	Stability, overlap, recurrence, locality, neighbor rank, or local coherence.
Regions and basins	Persistent regions, occupancy, boundaries, purity, recurrence, or hierarchical relationships.
Transport and paths	Accessibility, path length, path diversity, robustness, bottlenecks, transition corridors, or path entropy.
Recursive organization	Consistency across recursive indexing or abstraction levels.
Regime structure	Persistent structural regimes, transitions, excursions, recovery, or reentry.
Compression	Reduction in dimensions, rules, generator terms, paths, hierarchy, or computational burden while preserving structure.
Integrity	Preservation under controlled disassembly, perturbation, reconstruction, or scale reduction.
Transfer	Persistence of organization across time, datasets, domains, or independent observations.

9. Generator Compression

Generator compression asks what smaller or simpler generative organization can reproduce or preserve selected organization observed in a representation.

```

Observed organization S
|
Candidate generator G
|

```

```

Generate / simulate / reconstruct  $S_{hat}$ 
|
Compare( $S$ ,  $S_{hat}$ )
|
Complexity( $G$ )
|
Generator Compression Score

```

One general objective can be written as:

$$J(G) = L_{structure}(S, S_{hat_G}) + \lambda * Complexity(G)$$

where $L_{structure}$ is a structural loss and Complexity(G) may measure rule count, description length, parameters, dimensions, active relations, recursive levels, runtime cost, memory, or another compactness quantity. The equation is illustrative; no particular scalar objective is required.

9.1 Structural preservation targets

- Preserve neighborhood identity or neighbor ordering.
- Preserve path accessibility or transport costs.
- Preserve bottleneck structure.
- Preserve recursive hierarchy.
- Preserve structural regime boundaries.
- Preserve recovery or reentry pathways.
- Preserve predictive utility where a downstream task is included.
- Preserve independent qualification status or entitlement.

9.2 Generator compression versus ordinary compression

Ordinary compression may preserve the same data with fewer bits or dimensions. Generator compression instead seeks a compact generative organization capable of reproducing the relationships, transformations, pathways, or structural possibilities that matter for the intended purpose. The two forms of compression can be combined but are not identical.

10. Representation Comparison and Improvement Criteria

Candidate representations may be compared using a vector of measurements:

```

ScoreVector(R) = [
    structural_exposure,
    transport_robustness,
    integrity,
    transfer,
    predictive_utility,
    complexity,
    latency,
    memory,
    qualification_status
]

```

The system preferably does not require all quantities to be collapsed into a single score. A candidate can be retained if it is non-dominated, within an entitlement region, or preferred for a particular purpose.

10.1 Improvement conditions

- Greater structural exposure with equal or lower complexity.
- Equivalent structural support with reduced computational cost.

- Improved structural integrity with equivalent predictive performance.
- Improved predictive performance while preserving or increasing qualification status.
- Improved transfer across independent observations.
- Greater ability to detect structural transition, failure, recovery, or abstention conditions.
- Equivalent scientific entitlement with lower latency, memory, energy, or implementation complexity.

11. Representation-Improvement Algorithm

INPUT:

```

observations O
initial representation R0
curated generator set G
structural metrics M
optional task metrics P
engineering policy E
qualification policy Q

```

for cycle $t = 0 \dots T$:

1. diagnose limitations of R_t
2. select one or more candidate generators from G
3. construct candidate representations $R_{t,j}$
4. measure structural evidence $M(R_{t,j})$
5. optionally measure task evidence $P(R_{t,j})$
6. characterize generator complexity $C(G_j)$
7. screen candidates under engineering rules
8. freeze selected candidates
9. independently qualify candidates when required
10. determine entitled candidate set
11. apply engineering selection policy
12. designate selected representation as R_{t+1}
13. record lineage and evidence
14. stop or repeat according to predefined rule

12. Relationship to Independent Qualification and Entitlement

Representation improvement may generate a candidate that appears superior during engineering. That engineering result does not automatically establish scientific entitlement. In a preferred architecture, candidates selected for scientific or high-consequence use are frozen and evaluated by an independent qualification subsystem.

Engineering evidence

- > candidate proposal
- > immutable representation product
- > independent qualification
- > entitlement region / supported set
- > engineering selection among supported candidates

This separation prevents the representation-improvement loop from tuning directly to the same qualification evidence used to justify scientific support.

13. Generator Entitlement

A generator may be assigned an explanatory, predictive, or operational scope. The system may determine whether the evidence supports that scope.

- A generator may reproduce neighborhood structure but not path organization.
- A generator may reproduce one domain but fail to transfer to an independent system.
- A generator may improve prediction but fail structural qualification.
- A generator may explain a local regime while not supporting global generalization.
- A generator may be entitled to generate candidate representations without being entitled as a scientific mechanism.

Generator entitlement can therefore be represented as a mapping between demonstrated capacities and permitted claims rather than as a binary statement that the generator is correct.

```
GeneratorEntitlement {  
    supported_structures[]  
    supported_domains[]  
    supported_scales[]  
    predictive_support  
    explanatory_support  
    operational_support  
    limitations[]  
    qualification_refs[]  
}
```

14. Recursive Representation Improvement and Discovery

Each completed cycle may generate a new question about the representation. Examples include:

- Which structural relationships are still unexplained?
- Which component contributes disproportionate structural influence?
- Can the same organization be generated with fewer rules?
- Does an alternative scale expose more stable transport?
- Can the candidate transfer across an independent dataset or system?
- Does a new representation restore entitlement in a regime where a prior representation failed?

The next cycle may therefore use evidence from the prior cycle to propose a new bounded representation experiment. Human scientific authority may be required before the cycle changes generator families, accesses reserved data, publishes a claim, or initiates intervention.

15. Representation Improvement Record and Lineage

```
RepresentationImprovementRecord {  
    cycle_id  
    parent_representation_id  
    source_data_refs[]  
    candidate_generator_ids[]  
    candidate_representation_ids[]  
    structural_measurements[]  
    task_measurements[]  
    generator_complexity[]  
    qualification_refs[]  
    entitlement_region_ref  
    engineering_policy
```

```

    selected_representation_id
    rejected_or_deferred_candidates[]
    limitations[]
    next_question_or_stop_reason
    provenance[]
}

```

The record permits later investigators to determine why a representation changed and whether the change reflected engineering preference, scientific evidence, or both.

16. Example AIIA / RIX / SII / PIX / RSD Embodiment

```

PIX / representation runtime
  materializes and snapshots candidate views
  |
RIX / structural computation
  measures neighborhoods, basins, paths, topology, stability
  |
AIIA Platform / representation engineering
  generates, diagnoses, and compares candidates
  |
SII / independent qualification
  qualifies immutable candidates and constructs entitlement
  |
RSD / recursive scientific discovery
  uses evidence to propose the next authorized investigation

```

The named architecture is a non-limiting embodiment. The invention can operate with any representation runtime, structural computation engine, engineering platform, qualification system, or discovery controller that performs corresponding functions.

17. Early Enabled Generator Embodiments

To avoid requiring speculative autonomous generator search, an enabled early embodiment uses a curated generator catalog.

Generator family	Example construction
Direct representation	Use original or standardized coordinates/features.
Neighborhood representation	Construct local adjacency from distance, similarity, compatibility, or admissibility.
Constraint-weighted representation	Modify relationships according to domain or learned constraints.
Recursive representation	Construct higher-order regions from prior structural objects.
Transport representation	Represent path accessibility, path cost, or shortest/robust routes.
Basin / region representation	Aggregate observations into persistent structural

neighborhoods.

Multiscale representation

Maintain multiple predefined scales or hierarchy levels.

Compressed representation

Remove or aggregate low-value components subject to structural-preservation constraints.

18. Example Applications

Application	Illustrative use
Protein representations	Compare direct, biochemical, alignment-aware, relational, transport, and recursive representations and preserve the lineage of which representation exposes transferable structural organization.
World models	Generate alternative latent structural views and compare their ability to expose regime transitions, excursions, recovery, or failure.
Pharmaceutical AI	Improve representations used for molecular, biological, clinical, or manufacturing decisions while preserving independent qualification before deployment.
Robotics	Compare state representations according to reachable regimes, path robustness, recovery structure, and computational cost.
Scientific discovery	Iteratively refine organizational atlases or structural models as new evidence identifies unexplained structure.
Enterprise AI	Generate alternate structural views around an existing model without replacing the predictive model and select a view according to trust, cost, and use requirements.
Reasoning systems	Compress observed reasoning or transition histories into smaller reusable generator structures and test whether those structures preserve reachable conclusions or prediction quality.

19. Example Controlled Representation-Improvement Protocol

1. Freeze the source observations, candidate generator family, structural metrics, and engineering comparison policy.

2. Construct the reference representation.
3. Construct one or more candidate representations using bounded generator differences.
4. Measure structural organization without accessing reserved downstream test outcomes where required by study design.
5. Apply matched controls to distinguish generator structure from generic capacity or dimensionality effects.
6. Freeze candidate outputs and engineering evidence.
7. Submit predefined candidates for independent qualification.
8. Construct the supported or entitled candidate set.
9. Apply the predeclared engineering selection policy to the supported set.
10. Record selected representation, alternatives, evidence, limitations, and next-question rationale.

20. Detailed Algorithmic Embodiments

20.1 Algorithm A — Candidate representation generation

```

for generator  $G_i$  in curated registry:
    verify input compatibility
    instantiate frozen parameters
    construct candidate  $R_i$ 
    compute construction provenance
    store candidate for structural analysis

```

20.2 Algorithm B — Generator compression

```

INPUT: observed structural object  $S$ , generator family  $G$ 
for candidate generator  $g$  in  $G$ :
    construct or simulate  $S_{hat_g}$ 
    compute structural loss  $L(S, S_{hat_g})$ 
    compute generator complexity  $C(g)$ 
retain candidates satisfying structural preservation threshold
identify lower-complexity candidates or Pareto set
OUTPUT: compressed generator candidates

```

20.3 Algorithm C — Representation-improvement decision

```

INPUT: candidate profiles  $V_i$ , qualification  $Q_i$ , policy  $P$ 
1. discard candidates failing mandatory engineering or qualification conditions
2. identify supported candidate set or entitlement region  $E$ 
3. compare structure, utility, complexity, and cost within  $E$ 
4. apply policy  $P$ 
5. select  $R^*$ 
6. record why  $R^*$  was selected and which candidates remained scientifically supported

```

20.4 Algorithm D — Next-representation proposal

```

INPUT: selected representation  $R_t$ , diagnostic evidence  $D_t$ 
identify structural limitation  $L_t$ 
map  $L_t$  to eligible bounded generator modifications
rank modifications by expected information gain, structural need, or cost
propose one or more candidate generators for next authorized cycle
do not automatically claim scientific improvement until requalified

```

21. Controls, Reproducibility, and Scientific Discipline

- Separate exploratory candidate generation from confirmatory qualification.
- Freeze candidate-generator definitions or record every adaptive change.
- Do not use held-out or reserved outcomes to choose representation generators when those outcomes are intended for later confirmation.
- Use complexity-matched or shuffled controls where a structural improvement could arise from added capacity alone.
- Preserve negative candidate results and failed qualification outcomes in the improvement lineage.
- Do not infer that a generator is causal merely because it compresses or reproduces structure.
- Distinguish a generator useful for engineering from a generator entitled as a scientific explanation.
- When multiple candidates are equivalent within evidence tolerance, preserve the full entitled set rather than declaring the highest point estimate uniquely superior.
- Require human authorization for transitions from representation engineering to publication, intervention, or other high-consequence use when policy requires it.

22. Alternative Embodiments and Variations

- Candidate generators may be manually authored, retrieved from a registry, generated by rule composition, suggested by an AI system, or produced by optimization, provided the search space or provenance is recorded.
- The comparison process may be single-cycle or recursive.
- Structural evidence may be phenotype-blind, target-blind, supervised, unsupervised, or hybrid depending on the experiment.
- Generator compression may preserve only selected structural properties rather than the entire representation.
- Complexity may be measured by description length, parameter count, dimension, rule count, compute time, memory, energy, recursive depth, edge count, path count, or another resource.
- Representation improvement may select a more complex representation when the structural or operational gain justifies the added complexity.
- An improvement system may abstain from selecting a unique representation when evidence does not support one.
- Multiple parent representations may be combined to create a child representation.
- A representation may be decomposed and recombined into generator modules.
- The system may use trade-secret generator libraries while publishing only resulting qualified representation products.

23. Technical Advantages

- Treats representation improvement as a persistent computational capability rather than a one-time modeling decision.
- Separates the representation from the discovery process that produced it.
- Provides explicit structural criteria for comparing representations that encode the same observations.
- Supports compression that preserves generator-relevant organization rather than only coordinate reconstruction.
- Allows simpler representations to be preferred when scientific support is equivalent.
- Prevents representation engineering from automatically conferring scientific entitlement.
- Preserves full improvement lineage and failed alternatives for reproducibility.

- Supports bounded early implementations without requiring speculative autonomous generator discovery.
- Can integrate with existing numerical and predictive systems rather than replacing them.

24. Illustrative Claim-Style Invention Statements

1. A computer-implemented method comprising receiving observations; generating a plurality of candidate representations from the observations using different representation generators; measuring at least one structural property of each candidate representation; comparing the candidate representations using the structural property; and selecting or proposing a representation based at least in part on the comparison.
2. The method above wherein the structural property comprises neighborhood stability, basin organization, path accessibility, transport, bottleneck structure, recursive consistency, structural entropy, regime stability, excursion, recovery, reentry, or representation integrity.
3. A method further comprising computing a complexity measure for each representation generator and selecting a candidate that preserves a structural property while reducing generator complexity.
4. A computer-implemented method comprising receiving an observed structural organization; generating a candidate structural organization using a generator; comparing the observed and generated structural organizations; and determining generator compression from structural agreement and generator complexity.
5. The method above wherein generator complexity comprises description length, rule count, parameter count, dimension, active relation count, recursive depth, runtime cost, memory, or energy.
6. A method wherein candidate representations are generated from a curated generator registry rather than an unconstrained autonomous generator search.
7. A method wherein candidate generator parameters are varied only within predefined bounds during an engineering cycle.
8. A representation-improvement system comprising a generator registry, candidate-construction engine, structural-analysis engine, generator-compression engine, comparison engine, and lineage recorder.
9. A method further comprising freezing one or more candidate representations and submitting the frozen candidates to an independent qualification subsystem before permitting scientific or high-consequence use.
10. A method further comprising constructing an entitlement region of qualified candidate representations and applying a separate engineering selection policy only to members of the entitlement region.
11. A method wherein a numerically higher-performing candidate is not declared scientifically preferable when its evidence does not distinguish it from another qualified candidate.
12. A method further comprising selecting a candidate based on minimum complexity, cost, latency, memory, or energy subject to structural qualification constraints.
13. A method comprising iteratively using evidence from a first representation-improvement cycle to select or propose a generator modification for a second representation-improvement cycle.
14. A method further comprising recording a parent-child representation lineage identifying generators, structural measurements, qualification records, and selection policies.
15. A method comprising determining generator entitlement by mapping demonstrated generator capacities to supported explanatory, predictive, or operational scopes.
16. A method wherein a generator is permitted for representation engineering but not entitled as a scientific explanation.
17. A method comprising comparing two representations of the same observations and determining that one representation exposes a structural relationship not sufficiently observable in the other.

- 18.** A method comprising selecting a compressed representation that preserves a predefined neighborhood, transport, bottleneck, hierarchy, or regime property while reducing representation complexity.
- 19.** A method comprising using target-blind structural measurements to screen candidate representations before accessing a reserved supervised test outcome.
- 20.** A method comprising retaining multiple non-dominated candidate representations when no single candidate is supported as superior across predefined structural and engineering criteria.
- 21.** A method comprising selecting a subsequent scientific or engineering question based on a structural limitation detected in a selected representation.
- 22.** A computer-readable medium storing instructions that cause one or more processors to perform any method described herein.
- 23.** A method wherein an AI system proposes candidate generator modifications but a separate authorization rule controls whether the proposed generator is executed, qualified, published, or used operationally.
- 24.** A method comprising preserving failed representation candidates and associated evidence rather than overwriting the improvement history.

25. Relationship to Existing Portfolio and New Matter

Earlier filed work relating to constraint-governed representations, recursive indexing, recursive compression, transformational information, coordinate-structure divergence, structural excursion/recovery, and structural entitlement provides background, component technologies, and operational embodiments for representation improvement. The July Generator Compression and Representation Improvement memorandum expressly documented generator compression, representation improvement, generator entitlement, and discovery-process systems as strategic and potential IP directions.

The present disclosure focuses on the engineering architecture connecting those ideas: candidate generator registries; iterative construction and comparison of alternative representations; generator compression based on structural preservation and complexity; explicit lineage; separation from independent qualification; and recursive proposal of subsequent bounded representation experiments.

Appendix A — Draft Figure Schematics

FIG. 1. Representation Improvement Loop

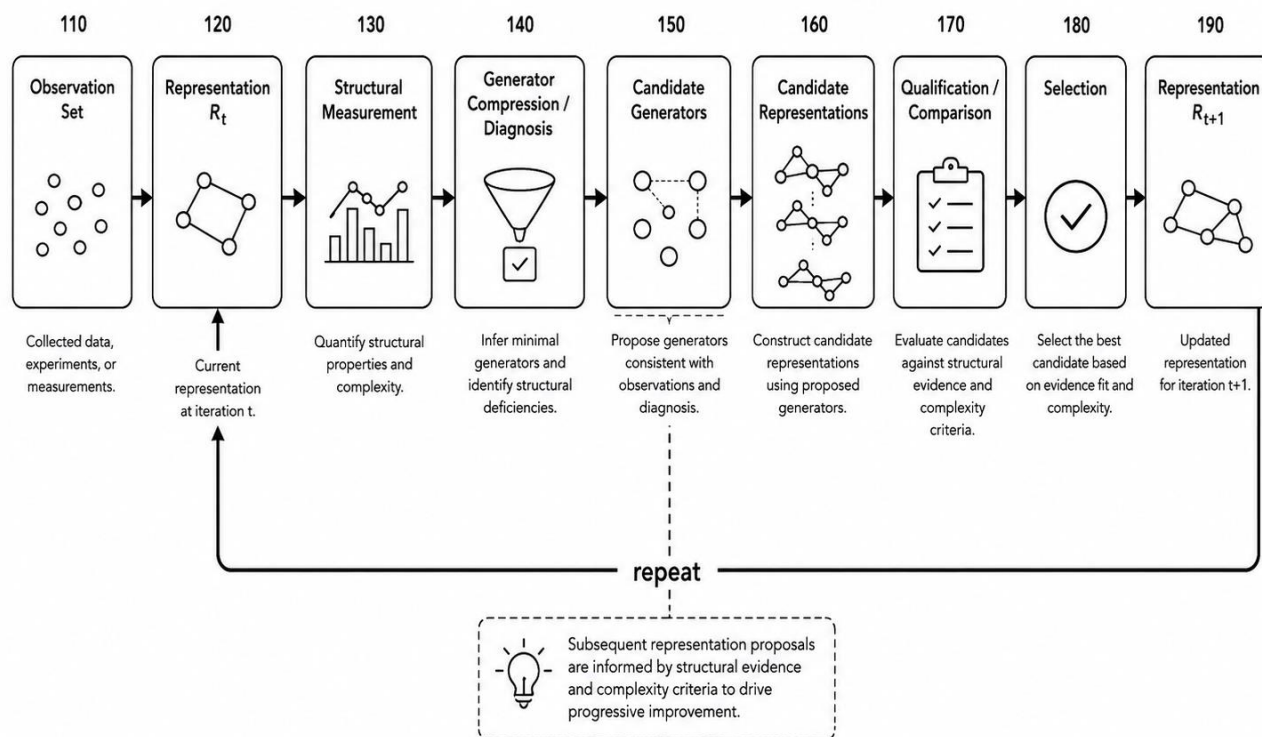


FIG. 1. Overall representation-improvement loop from observations through candidate generation, structural measurement, generator compression, comparison, selection, and subsequent representation.

FIG. 2. Candidate-Representation Generator Registry

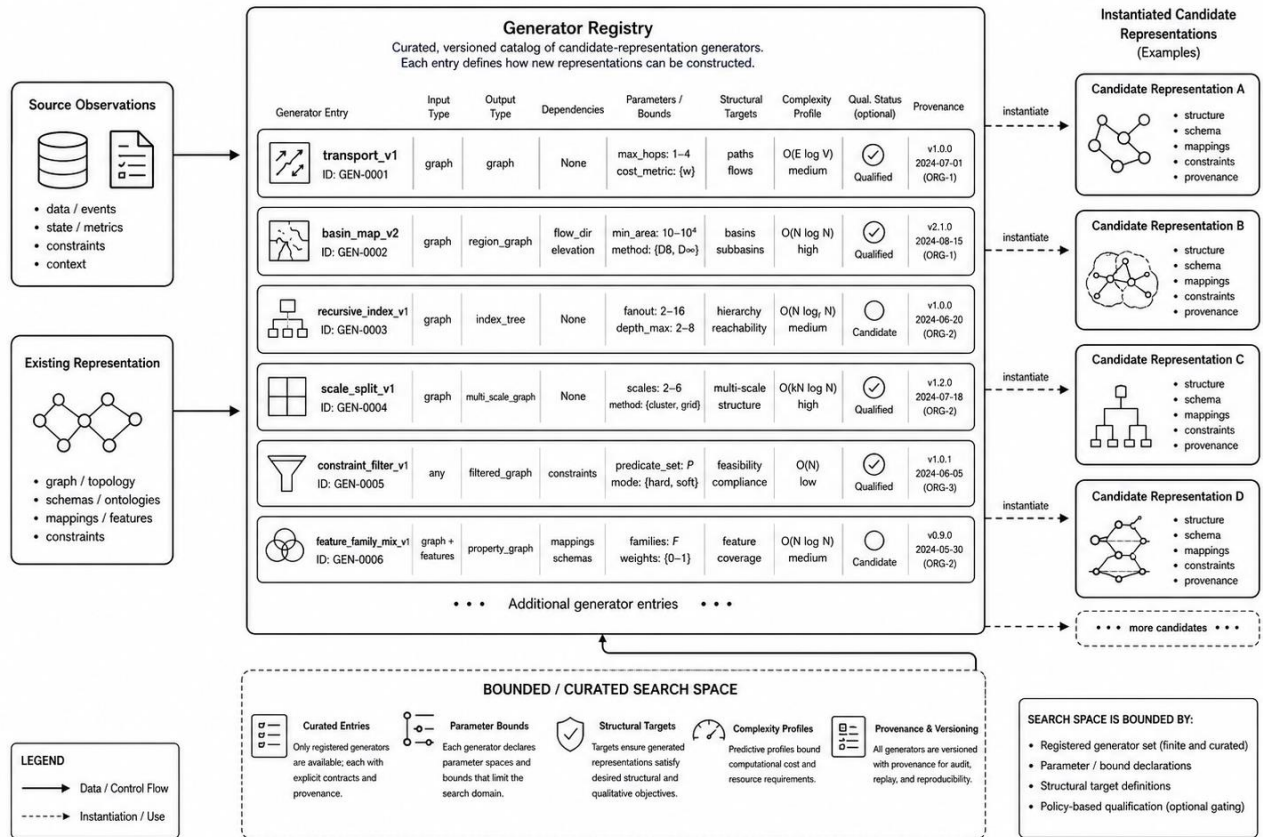


FIG. 2. Example candidate-representation generator registry.

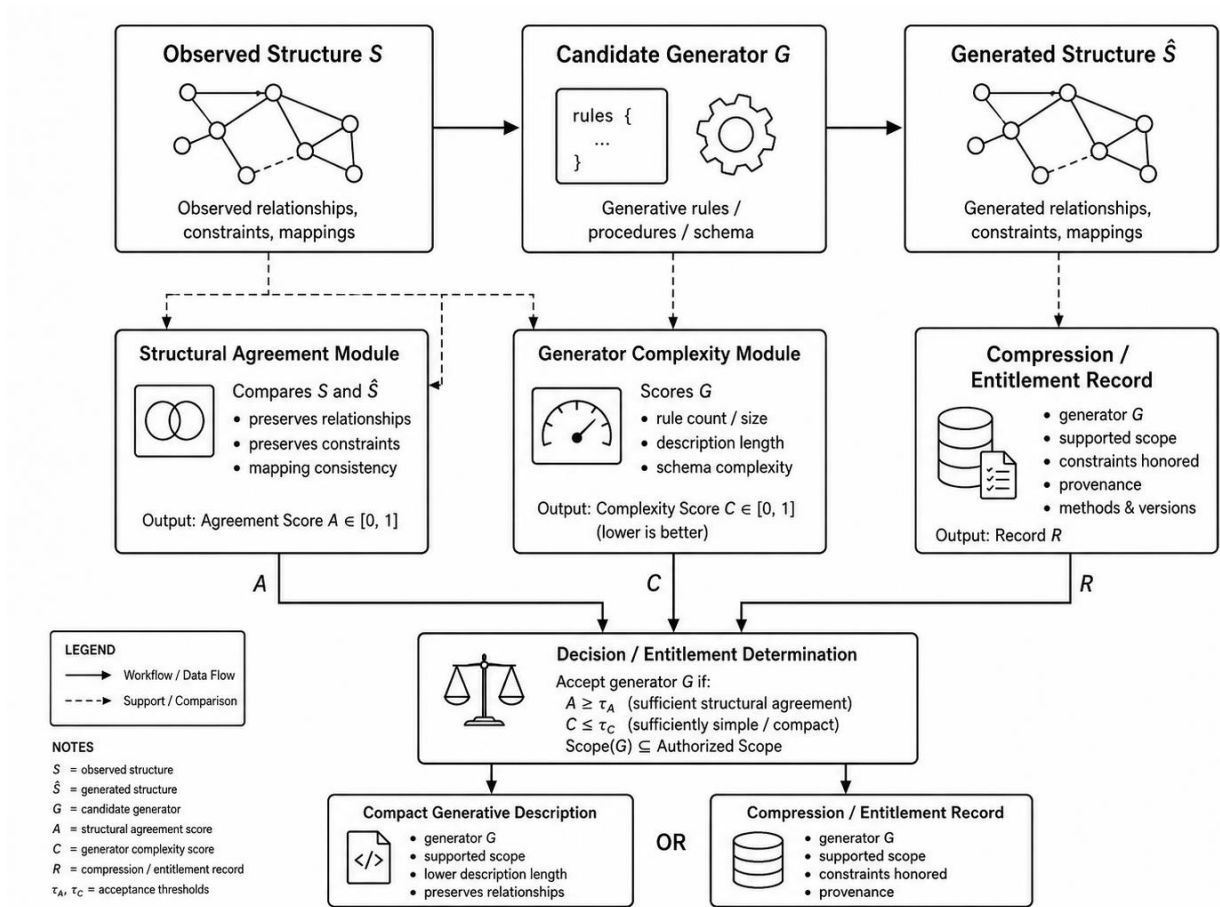


FIG. 3. Generator Compression Architecture

FIG. 3. Example generator-compression architecture converting observed organization into a compact generative description.

FIG. 4. Multidimensional Representation Comparison Profile

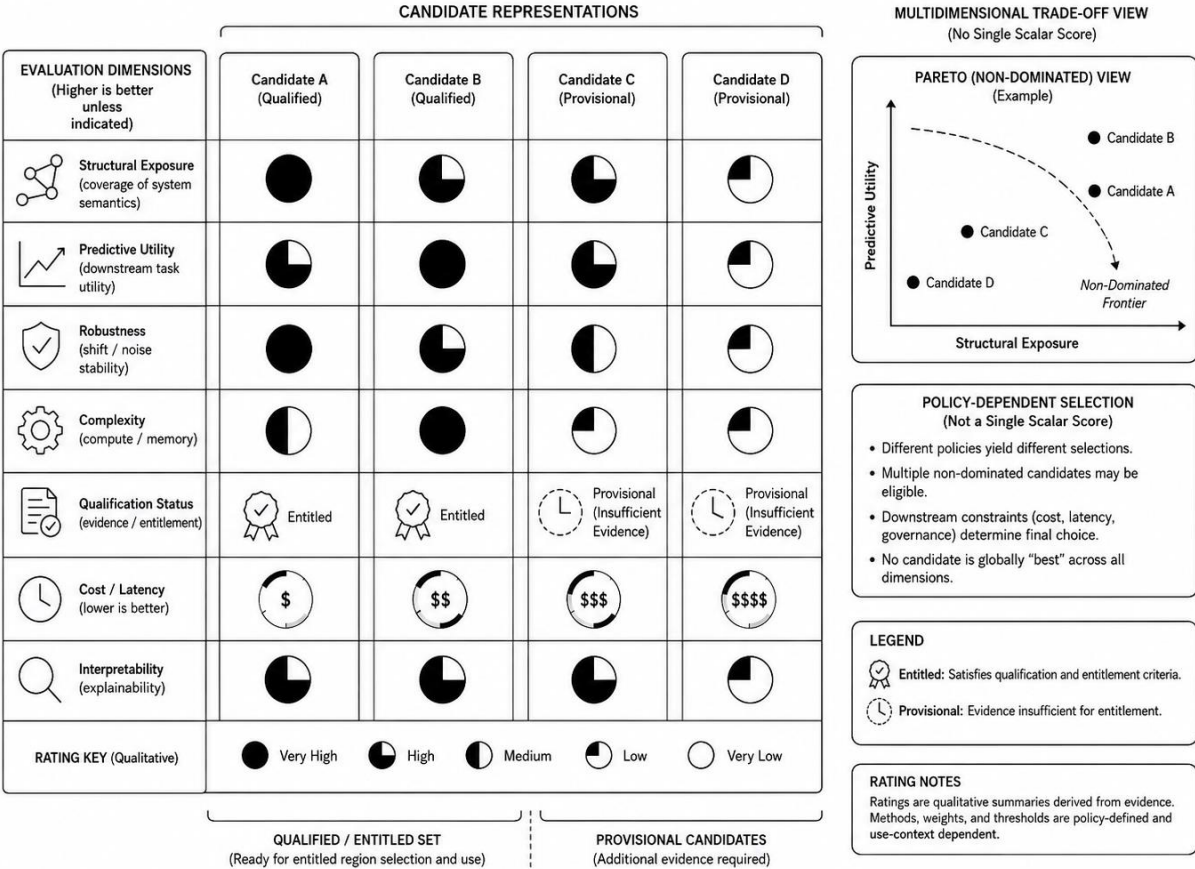


FIG. 4. Example multidimensional representation comparison profile.

FIG. 5. Entitlement Region and Engineering Selection

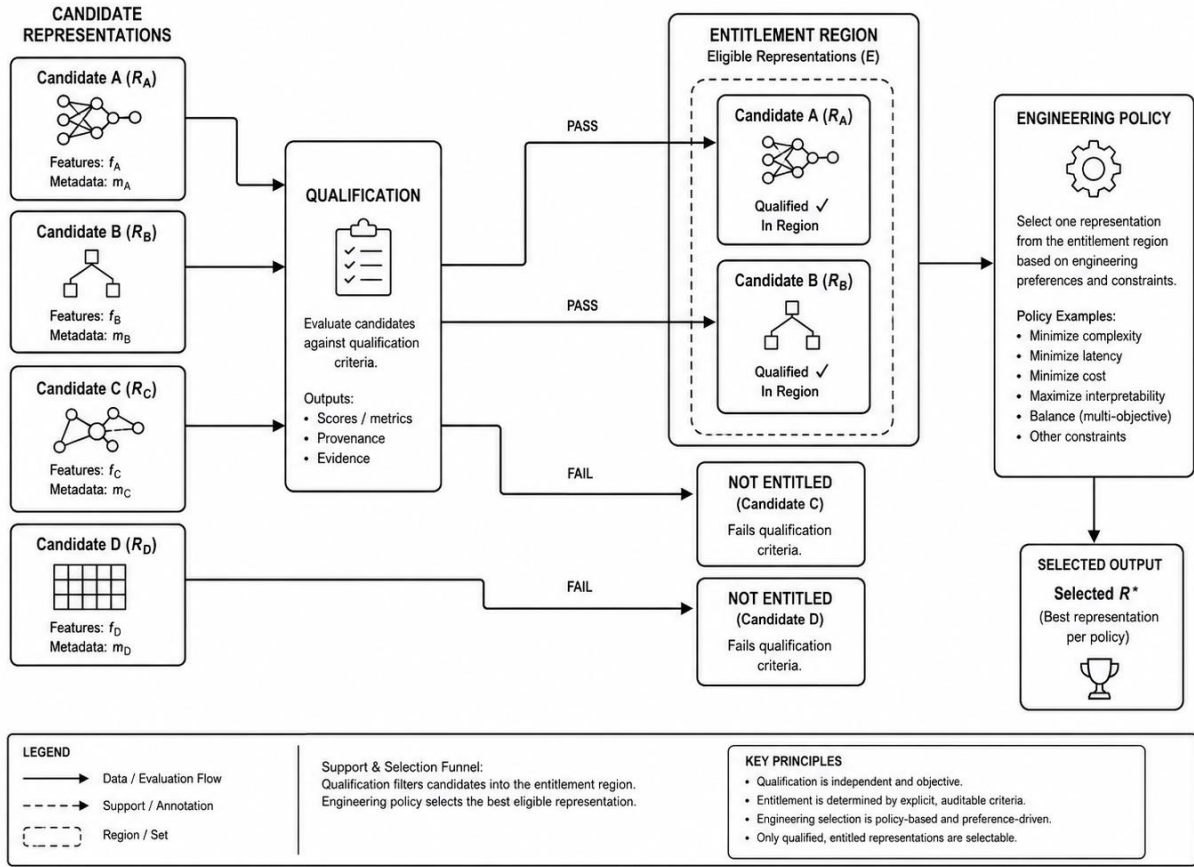


FIG. 5. Example entitlement region and engineering selection from qualified candidates.

FIG. 6. Recursive Representation-Improvement Lineage

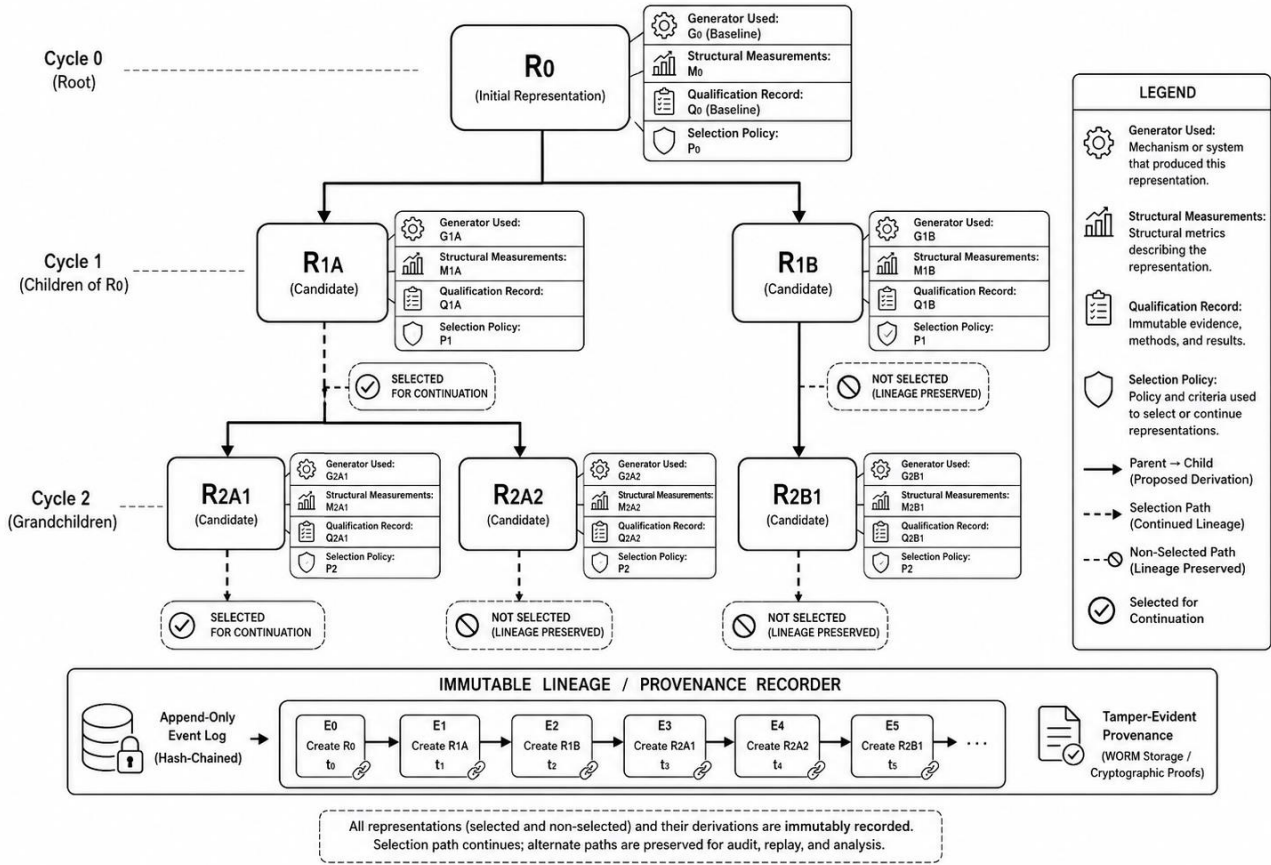


FIG. 6. Example recursive representation-improvement lineage.

FIG. 7. Separation Between Representation Engineering and Independent Qualification

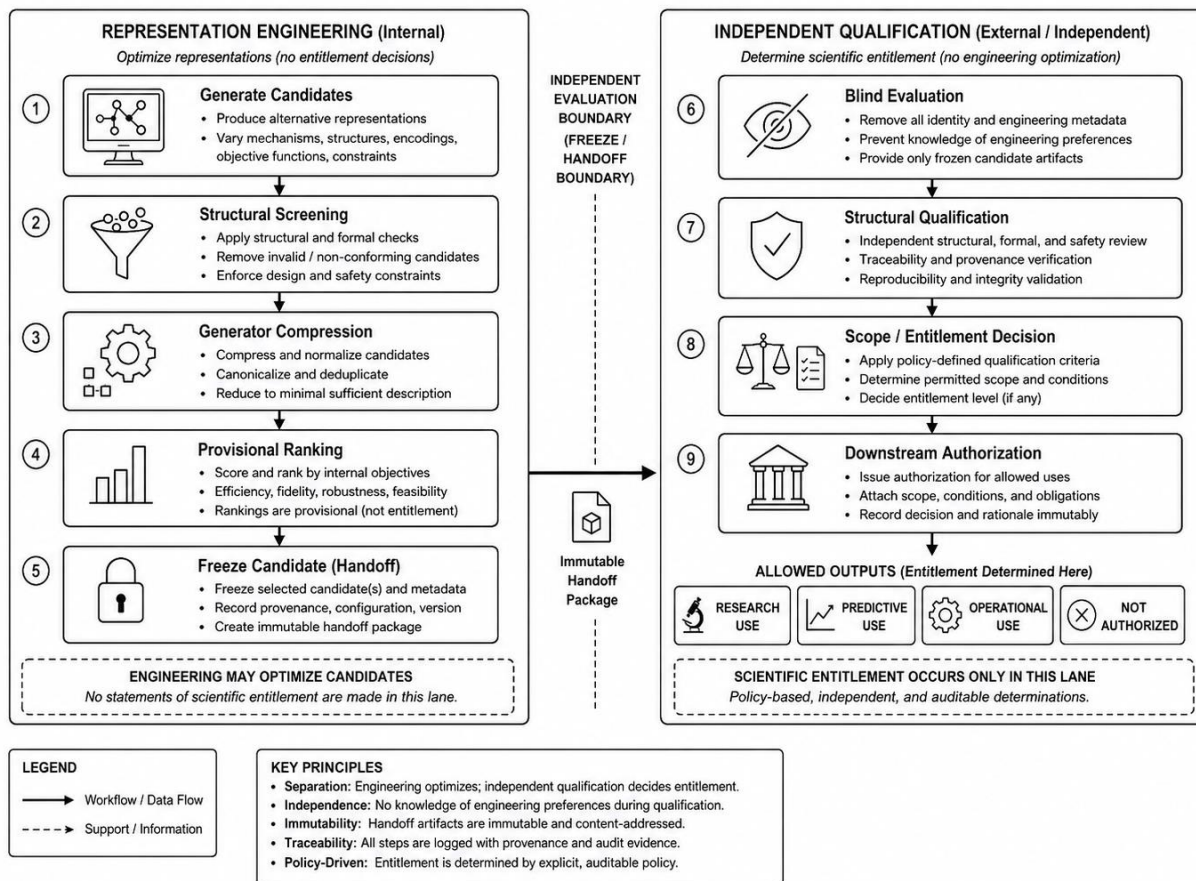


FIG. 7. Example separation between representation engineering and independent qualification.

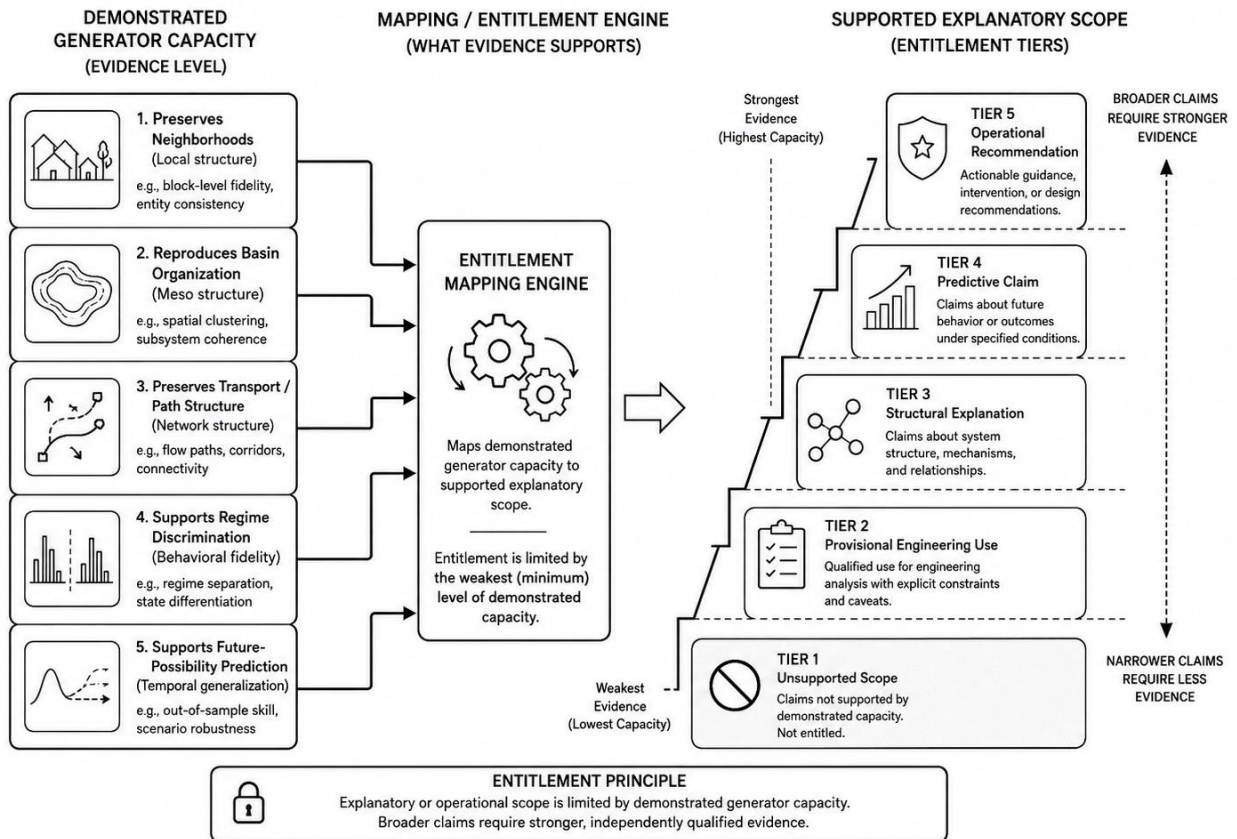


FIG. 8. Generator-Entitlement Architecture

FIG. 8. Example generator-entitlement architecture linking claimed explanatory scope to demonstrated generator capacity.

FIG. 9. Structural Measurements Used in Representation Comparison

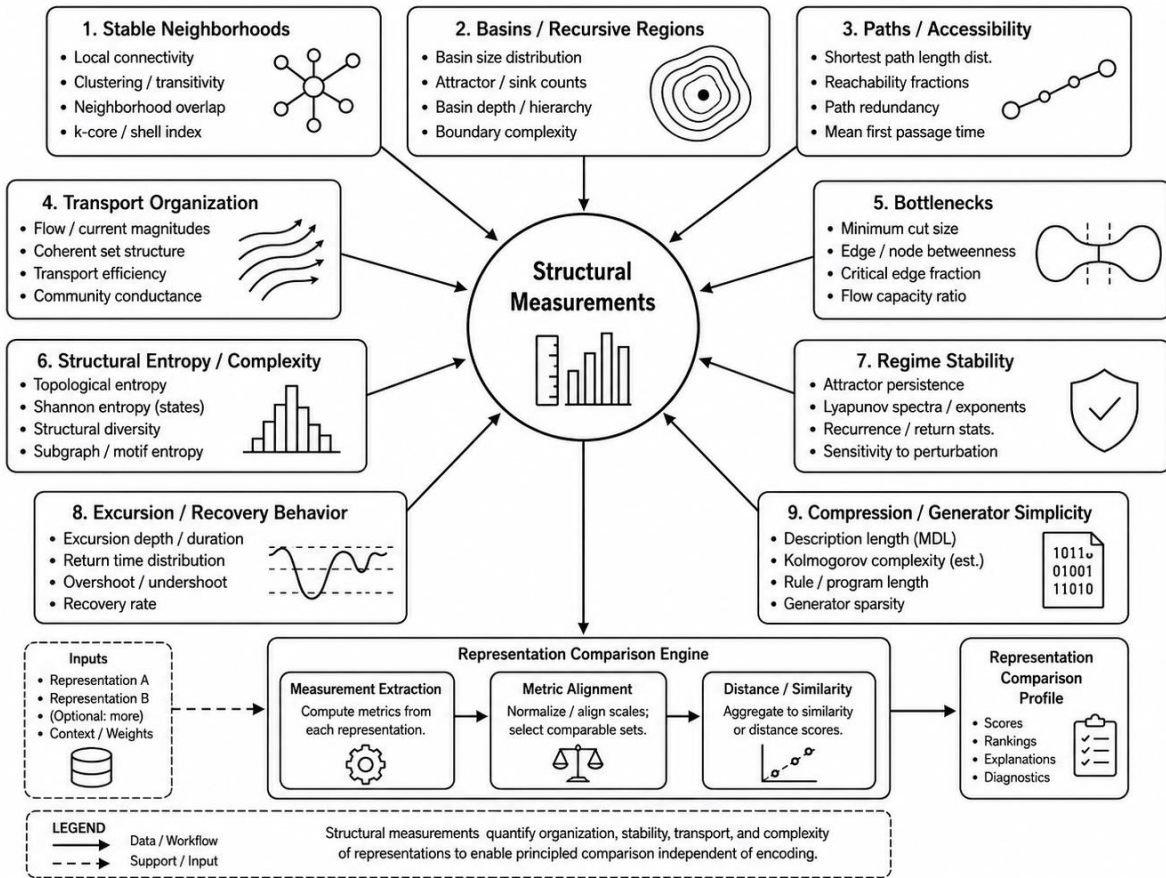


FIG. 9. Example structural measurements used in representation comparison.

FIG. 11. Example curated generator search space with bounded composition and parameter variation.

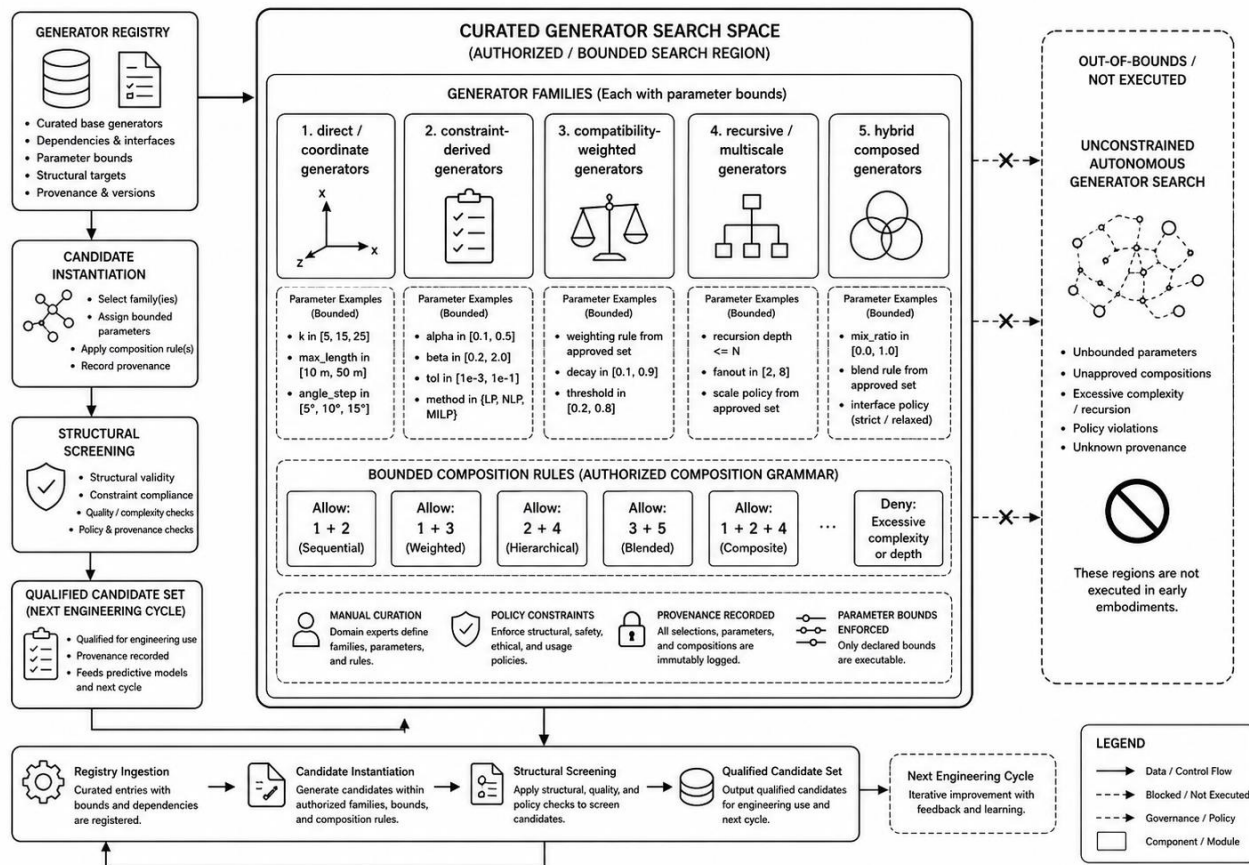


FIG. 10. Example implementation with a representation runtime, structural computation engine, qualification subsystem, and recursive discovery controller.

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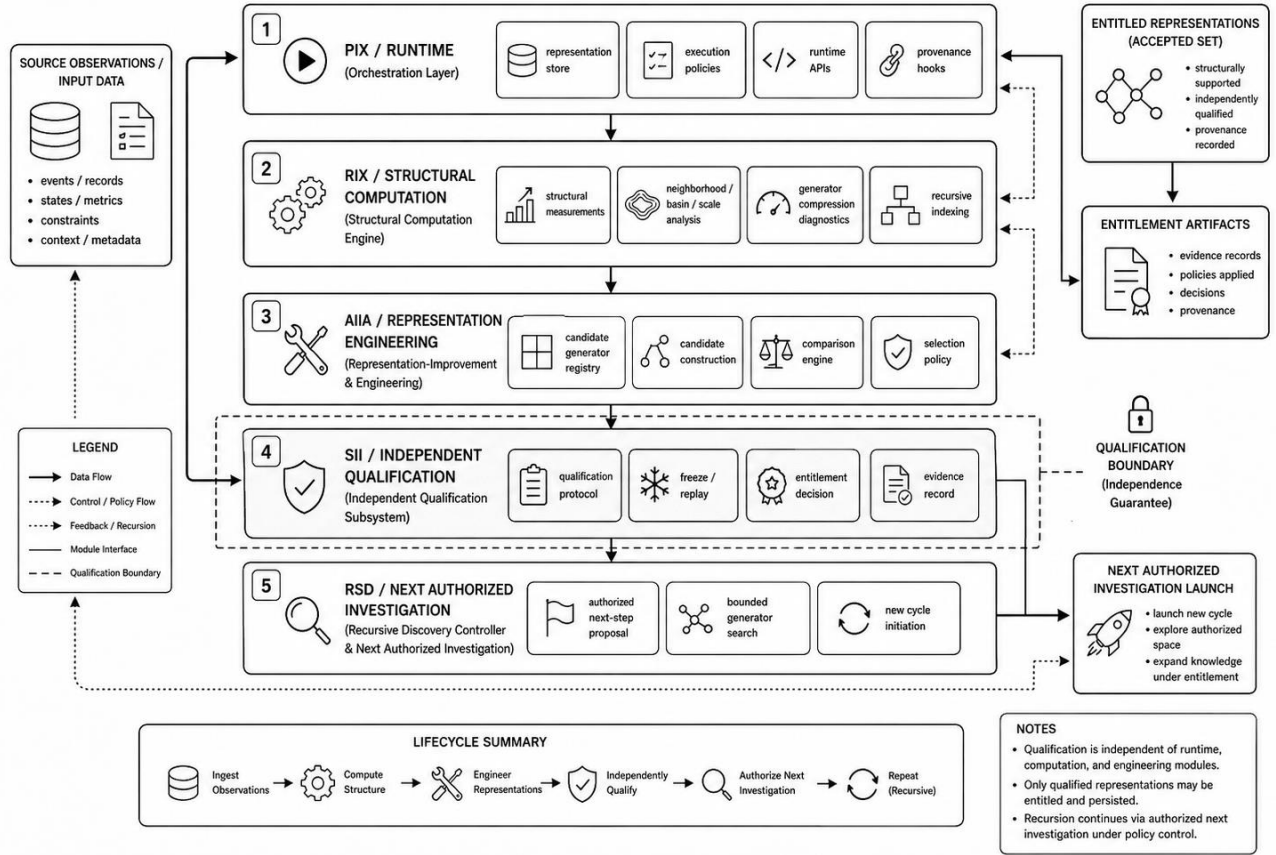


FIG. 11. Example curated generator search space with bounded composition and parameter variation.

FIG. 12. Example representation-improvement record stored with provenance and parent-child lineage.

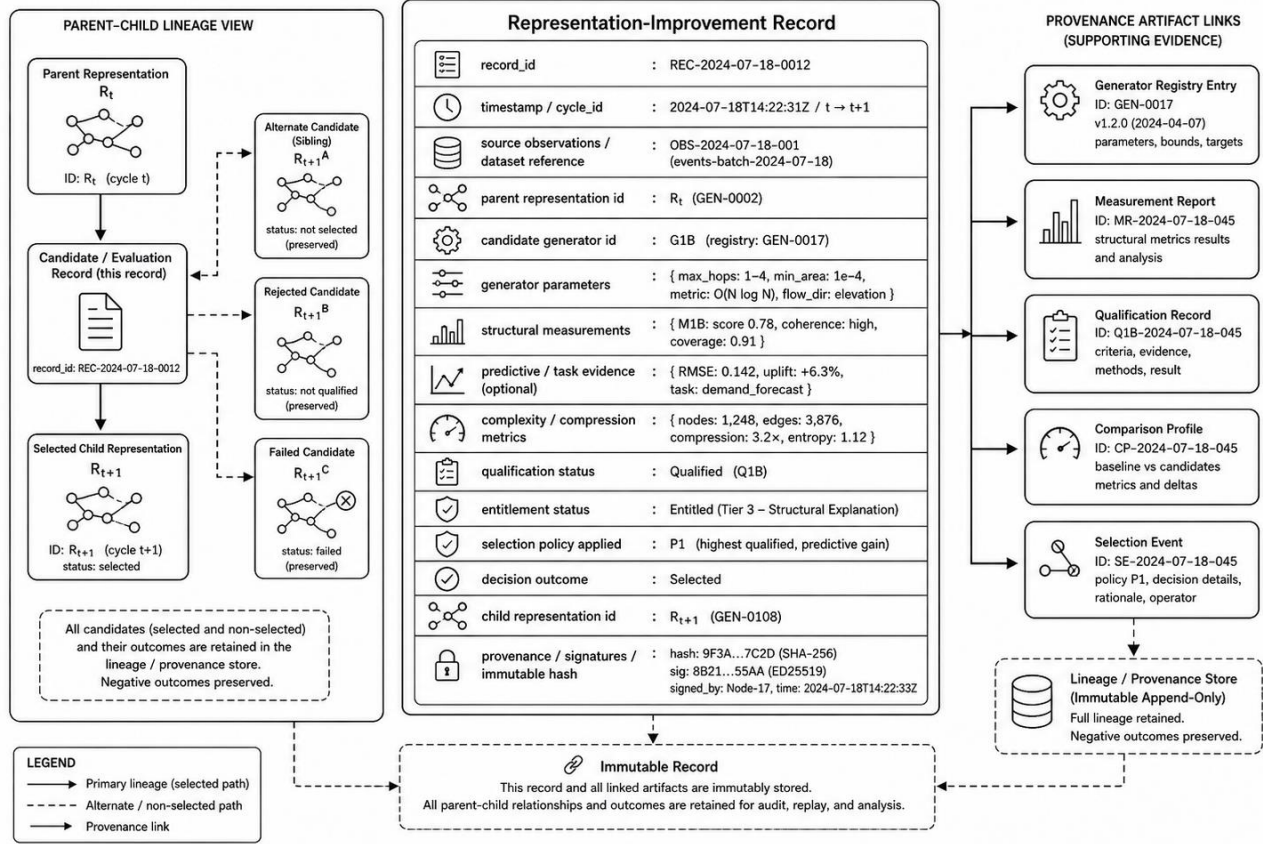


FIG. 12. Example representation-improvement record stored with provenance and parent-child lineage.

Appendix B — Example Generator Registry Schema

```

GeneratorRegistryEntry {
  id: "transport_v1"
  input_type: "direct_representation"
  output_type: "transport_representation"
  required_dependencies: ["neighborhood_index"]
  parameters: {k, scale, weighting_rule}
  parameter_bounds: predefined
  structural_targets: ["reachability", "path_robustness"]
  complexity_profile: estimated
  qualification_status: optional
  provenance: immutable
}

```